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The Impact of the COVID-19 Pandemic on the Behavior of Online Gig Workers

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Abstract. *Problem definition:* Using labor supply data from a large online education platform with more than 100,000 gig workers, we investigate how online gig workers changed their behavior after the outbreak of the coronavirus disease 2019 (COVID-19) pandemic and what drove the changes. *Methodology and Results:* Online gig workers sharply increased their labor supply on the platform by 23% from the announcement of national emergency to the end of April (stage 1); the increase became smaller in May and June (stage 2) and disappeared in July and August (stage 3). Year to year difference-in-difference analyses show that these findings are robust after controlling for seasonality and worker heterogeneity. *Results:* We show that the increase in gig workers' labor supply is not driven by a higher demand or excessive entry of new workers during the pandemic. A series of mediation analyses indicates that unemployment and nonpharmaceutical interventions (NPIs) rather than the risk of contracting COVID-19 can better explain why online gig workers increased their labor supply. The impact of unemployment is smaller than that of NPI policies, indicating that the increase in gig workers' labor supply is more driven by temporary changes in working arrangements because of the policies rather than relatively long-term changes in employment situations. We also examine how online gig workers change their quality of work and how their earning potential on the platform relates to their changes in behavior during the pandemic. *Managerial implications:* Our findings provide insights for the management of online gig workers during major disruptions, like the COVID-19 pandemic.

Supplemental Material: The online appendix is available at <https://doi.org/10.1287/msom.2022.1113>.

Keywords: gig economy • platform operations • worker behavior • personnel operations • COVID-19 pandemic

1. Introduction

With the development of digital technology, gig economy platforms, such as Uber, DoorDash, Upwork, and many others, are growing rapidly. They use digital platforms to connect freelancers with customers to provide short-term services, providing people with opportunities to work as independent contractors and with flexible schedules. The gig economy has already become an important sector in the economy, and gig workers also consist of a significant portion of the population. In 2017, 34% of the workforce (about 55 million people) in the United States was gig workers, and the number was predicted to be 43% in 2020.¹ Thus, understanding gig workers' behavior is crucial from a macro-level policy-making perspective.

Gig workers have significant differences compared with traditional employees. They work independently and can freely decide when and for how long they would like to work. The pay they get is directly determined by their amount of work and thus, can be

highly volatile. Many gig workers also have other full-time or part-time jobs besides working on the gig economy platforms (Chen et al. 2019, Guda and Subramanian 2019, Cook et al. 2021). Therefore, the management of gig workers is completely different from that of traditional workers, and understanding gig workers' behavior is important for the operations of gig economy platforms.

Starting in December 2019, a novel coronavirus disease, named the coronavirus disease 2019 (COVID-19), has spread around the world and changed many aspects of our lives. As the number of confirmed cases ramped up across the world and in the United States, the World Health Organization declared COVID-19 a pandemic on March 11, 2020,² and the U.S. Government announced a national state of emergency on March 13, 2020.³ After that, many state and local governments in the United States enacted nonpharmaceutical interventions (NPIs), such as "shelter-in-place" (SIP) orders, closures of day cares and schools, closures of restaurants and bars, etc.,

to restrict social and business activities so as to slow down the spread of the disease (Raifman et al. 2020). The shock brought by the pandemic as well as the NPI policies made many people unable to work on-site or even lose their jobs (Coibion et al. 2020, Gupta et al. 2020, Baek et al. 2021). In such a scenario, people can work more on gig economy platforms to remedy their loss of income.⁴ The closures of schools and day cares also required parents to spend more time caring for their family, and gig jobs became more appealing in this case because of their flexibility.⁵ At the same time, the risk of contracting COVID-19 may have also made workers more willing to work online.⁶ Thus, it is not surprising that the labor supply to gig economy platforms increased during the pandemic,⁷ especially on those gig economy platforms that allow workers to work online (like Upwork, Catalant, Guru.com, etc.). However, as the pandemic lasts, most states in the United States lifted the lockdown policies and gradually reopened businesses in May and June 2020 to save the economy. Following that, the unemployment rate gradually declined,⁸ but the numbers of new cases and new deaths of COVID-19 in the United States soared again and fluctuated afterward.⁹ Without knowing the key driving force of the increase in gig workers' labor supply, it is unclear whether and how long the increase in labor supply to online gig economy can persist. It is also unclear how gig workers changed their quality of labor supply during the pandemic and how gig economy platforms should have reacted to the changes in gig workers' behavior so as to better manage gig workers during the pandemic.

In this paper, we attempt to utilize the shock brought by the COVID-19 pandemic to understand how online gig workers changed their behavior during such a major disruptive event and what drives the changes. In particular, we investigate how they changed their quantity and quality of labor supply to the online gig economy platforms. We also try to disentangle the underlying mechanisms that drive the change in their labor supply to online gig economy.

Our main data set is from a large online education company that recruits native English speakers in the United States and provides an online platform for them to teach Chinese kids spoken English. The company is a gig economy platform that allows people to work at home as independent contractors. As of 2020, it has attracted over 100,000 people in the United States to work on the platform. The platform does not require prior teaching experience. In order to work on the platform, a potential worker only needs to apply on the platform and pass the interview and background check. After that, she will receive teaching guidance from the company and then, become a registered worker on the platform. A registered worker can open up half-hour-

long teaching slots on the platform, and each student on the platform can book teaching slots opened by any worker.¹⁰ Once a slot is booked, the matched worker and student will have a one-on-one English class session through an online classroom portal at the scheduled time. A worker will earn about 10 U.S. dollars for each finished class, which translates to an hourly wage of around 20 U.S. dollars. Notice that every registered worker can freely choose to open any number of slots ranging from 0 to 48 in each day, as long as she is available during the opened time slot. Once a worker opens a time slot and a student books it, the worker must meet with the student at that time slot; otherwise, the worker will be severely penalized. Thus, the number of time slots opened by a worker per day (*Daily Open Slots*) reflects the amount of labor she is able and willing to supply on the platform, which is a good indicator of labor supply. We only focus on the labor supply of existing gig workers on the platform. We abstract away from the entry of new gig workers because it is largely determined by the company's recruiting activity rather than the actual willingness to supply on the labor market.

Our data set has several unique advantages. First, unlike traditional labor data, which reflect the equilibrium outcome that is determined by both the supply side and the demand side, the key variable we observe (*Daily Open Slots*) reflects the willingness to work of gig workers that is not directly determined by the demand side. Second, studying the impact of major shocks (such as the COVID-19 pandemic) on labor supply is notoriously difficult because such shocks often affect both the supply and the demand of an economy at the same time. In this case, however, our data set provides a unique opportunity because the supply side and the demand side are geographically separated, and the shock of COVID-19 affected the demand side prior to it impacting the supply side. The demand side on this online platform is in China, where the COVID-19 pandemic happened earlier and had already been well controlled by the time the pandemic rose in the United States. (See details in Online Appendix A1.) Third, we have detailed worker-day-level and class-level data over a relatively longer time horizon in both 2019 and 2020, which allows us to comprehensively study the impact of the COVID-19 pandemic on gig workers' behavior and also, to test the underlying mechanisms.

We first analyze the impact of the COVID-19 shock using worker-day-level data from January to August 2020. We find that the average labor supply of gig workers (i.e., the average number of slots opened by a worker per day) on the platform increased sharply right after the declaration of the national emergency (March 13, 2020). From the day after the national emergency to the end of April (stage 1), the average labor supply was 23.2% higher than before the pandemic.

The increase became smaller in May and June (stage 2) and disappeared in July and August (stage 3). To account for seasonality, which may drive the change in labor supply, we use data from the same period in 2019 as a control group and perform year to year difference-in-difference (DiD) analysis (Bandiera et al. 2005). Our findings still hold after seasonality and worker heterogeneity have been controlled for.

We then test the possible mechanisms that drive the change in labor supply. We first show that the increase in gig workers' labor supply is not because of the competition brought by newly entered workers or a significant increase in demand. We find three main factors brought by the pandemic that may explain the increase in gig workers' labor supply to the online platform: the risk of contracting COVID-19, which makes them switch from offline to online jobs; the rise of unemployment; and the implementation of NPI policies, which make people unable to work on-site or prefer jobs with flexible working schedules. The variations in COVID-19 severity and unemployment rates across states and over time as well as the different starting and ending dates of NPIs across states allow us to causally estimate how much each of these factors can explain gig workers' change in labor supply. We conduct a series of mediation analyses, showing that among the three factors, NPI policies explain most of the increase in gig workers' labor supply. Gig workers increase their labor supply on the online platform not because of the direct risk of contracting COVID-19. Instead, the pandemic severity affects gig workers' labor supply through influencing the NPI policies and unemployment. The impact of unemployment is much smaller than that of NPI policies, indicating that the increase in gig workers' labor supply is more driven by temporary changes in working arrangements because of the policies rather than relatively long-term changes in employment situations. This is also consistent with the fact that gig workers' labor supply quickly gets lower in stage 3, when all work-related NPIs have ended.

We also analyze how gig workers' earning potential on the platform relates to their change in labor supply. A gig worker's earning potential on the platform can be measured by two factors: her hourly wage and the likelihood of her opened slots being booked by students. We find that gig workers with lower earning potential have a less stable labor supply; although gig workers all sharply increase their labor supply in the early stage of the pandemic, gig workers with lower earning potential are more likely to reduce their labor supply in the later stage when workers have more alternative working opportunities as the work-related NPIs end and the unemployment rate declines. This finding reinforces the conclusion that the increase in gig workers' labor supply during the pandemic is

mainly driven by the change in working arrangements and working opportunities rather than the risk of contracting COVID-19.

Lastly, we investigate how gig workers change their quality of labor supply on the platform as well as the relations between their quantity and quality of labor supply. We find that for both before and after the pandemic, when a gig worker provides more labor supply, her quality of labor supply tends to decrease, although this effect is smaller in stage 1 because of severe competition on the supply side. In stage 2 and stage 3, as the NPIs ended and the unemployment rate declined, gig workers have more other working options, and the supply side on the platform becomes less competitive; thus, individual gig workers on this platform tend to have worse performance (compared with before the pandemic). This is consistent with our findings regarding the key driving forces of gig workers' change in labor supply. However, high-quality workers tend to increase their labor supply more during the pandemic, especially in stage 2 and stage 3. As a result, we see better average performance measures of booked slots throughout the three stages.

Our findings can help gig economy platforms make better predictions about their supply side, which can be crucial for their operational plans when facing a pandemic or other major social disruptions that may happen in the future. We also provide insights about how gig workers' earning potential relates to their quantity of labor supply as well as the relations between quantity and quality of labor supply, which can help gig economy platforms better manage their labor supply—for example, to better design their incentive scheme. Our findings can also be valuable for policy makers, especially with regard to the impact of NPI policies on people's working arrangements and working opportunities.

The paper proceeds as follows. Section 1.1 reviews related literature and discusses our contributions to the literature. Section 2 provides details about the data sets we use. Section 3 investigates the impact of the COVID-19 pandemic on gig workers' labor supply. Section 4 investigates the key driving forces of gig workers' change in labor supply. Section 5 analyzes the relationship between gig workers' earning potential on the platform and their change in labor supply, and Section 6 investigates how gig workers change their quality of labor supply as well as the relationship between quantity and quality of labor supply. Section 7 discusses the managerial and policy implications of our findings, the limitations of the paper, and future research directions.

1.1. Literature Review

First, our paper is related to the recent literature discussing the impact of the COVID-19 shock on the

labor market. One stream of the literature studies the impact of lockdown policies on the labor market. For example, Forsythe et al. (2020) find that the economic collapse, reflected by the decrease in job vacancies and increase in unemployment, was not caused solely by the stay-at-home orders, and therefore, lifting these orders is unlikely to solve the problem. Similarly, Aum et al. (2021) and Kong and Prinz (2020) find that most of the short-run increase in unemployment during the pandemic was not because of the shutdown policies. Another stream of literature studies the unequal impact of the pandemic and arrives at a similar conclusion: that female workers, less educated workers, and minorities experienced a larger decline in the likelihood of full-time work and work hours during the pandemic (e.g., Adams-Prassl et al. 2020, Alon et al. 2020, Cowan 2020, Montenegro et al. 2020, Papanikolaou and Schmidt 2022). These papers mostly rely on self-reported survey data and study the impact of COVID-19 on the general labor market. To the best of our knowledge, our paper is the first attempt to use proprietary data to study the impact of the COVID-19 pandemic on gig workers. As we have discussed in Section 1, the unique advantages of our data allow us to perform a comprehensive study of the impact of the COVID-19 pandemic on online gig workers' behavior.

Our paper also contributes to the literature of gig economy platforms and platform operations. From a macro level, existing literature has shown that a higher unemployment rate can lead to an increase in labor supply to gig economy (Katz and Krueger 2017, Borchert et al. 2018, Jackson 2020, Huang et al. 2020). From a micro level, the existing literature has studied various mechanisms that could affect platforms' labor supply and operational efficiency, such as dynamic pricing and incentive schemes (Cachon et al. 2017), wage incentive (Chen et al. 2022), queuing control (Taylor 2018), matching demand and supply (Kabra et al. 2020), information design (Cui et al. 2020, Mejia and Parker 2021, Xu et al. 2021), and assortment recommendation (Feldman et al. 2022).

Our paper is also related to the literature of personnel operations. The literature of personnel operations studies various factors that can influence workers' performance and how to improve workers' performance. For example, Huckman et al. (2009) investigates how team familiarity and individual role experience affect workers' performance in a team. Tan and Netessine (2014) studies the impact of workload on worker's performance. Buell and Norton (2011) and Buell et al. (2017) study how operational transparency could improve workers' motivation and in turn, the work quality. Song et al. (2018) finds that public disclosure of relative performance feedback, along with the validation of the best practices being shared, can improve

worker productivity. Staats et al. (2011) finds that implementing a lean production system improves the knowledge production efficiency, and Gardner et al. (2012) studies how teams can develop a knowledge integration capability to dynamically integrate members' resources into higher performance. We study the impact of a major social disruption—the COVID-19 pandemic—on gig workers' behavior. Our results provide operational implications for gig economy platforms to better predict and manage their labor supply in major social disruptions.

2. Data

2.1. Labor Supply Data

As mentioned earlier, our main data set is from a large online education platform that has more than 100,000 gig workers in the United States. Because of data confidentiality, we randomly sample 40,000 workers from the workers who have opened at least one teaching slot during January to August 2020. As the entry of new workers is largely determined by the company's recruiting activity, we restrict our sample to the workers who registered on the platform before January 1, 2020 to focus on existing workers on the platform. To perform year to year DiD analysis, we also randomly sample the same number of workers who have opened teaching slots during January to August 2019 (and entered the platform before January 1, 2019)¹¹ and will treat the 2019 sample as the control group in the year to year DiD analysis.

Considering the weekly seasonality of labor supply, we choose January 5, 2020, the first Sunday in 2020, as the first date in the 2020 sample and keep 34 weeks in the data (the last day in the 2020 sample is then August 29, 2020). We also choose the first Sunday, January 6, 2019, as the first date in the 2019 sample and keep the same number of days in the data. We define "day of year" as the number of days between the current date and the first date in the sample and correspond each day in 2019 to the same day of year in 2020 when comparing the two years, so that the comparison is not affected by the weekly seasonality. When controlling for day fixed effects in the regression analyses, we are in fact controlling for day of year fixed effects.

Our main data set is at the worker-day level. The key variable that measures a gig worker's quantity of labor supply is the number of slots opened by the gig worker in a day (*Daily Open Slots*). We consider *Daily Open Slots* as a proper measure of a gig worker's labor supply for several reasons. First, a gig worker can open any number (between 0 and 48) of slots as long as she is able to and willing to work for that time slot, regardless of demand. This is in contrast to traditional employment data, which reflect the equilibrium outcome that is

determined by both the supply side and the demand side. Second, once the worker has opened a slot, she is indicating that she is available for the time slot, and students can freely book the slot. Once a student has booked the slot, the worker cannot cancel or reschedule the class freely; otherwise, the worker may get a monetary penalty, and her “class completion rate,” an important performance measure of workers on the platform, will also be negatively affected. (According to our data, only 0.84% of booked slots are cancelled or rescheduled by workers.) Thus, workers will open slots only if they believe that they will be available for the time slots and they are willing to work during the time slots. Therefore, *Daily Open Slots* is a fairly accurate measure of the labor supply that reflects gig workers’ willingness and ability to supply on the platform.

We also collected class-level data to measure gig workers’ performances in booked classes, reflecting their quality of labor supply. The detailed descriptions of gig workers’ performance measures are provided in Section 6.

Table A1(a) in the online appendix provides the definitions of variables in the labor supply data, and the summary statistics of these variables are available in Table A1(b) in the online appendix. Some variables have been normalized because of data confidentiality. See the details in Table A1(a) in the online appendix.

2.2. Epidemiological Data, Unemployment, and Policy Interventions

To measure the severity of the COVID-19 pandemic, we collect the following epidemiological data: new cases per 1,000 people, new deaths per 1,000 people, test positive rate (equal to the daily positive tests/daily total tests), COVID-19 inpatients per 1,000 people, COVID-19 inpatient beds utilization rate, and mortality

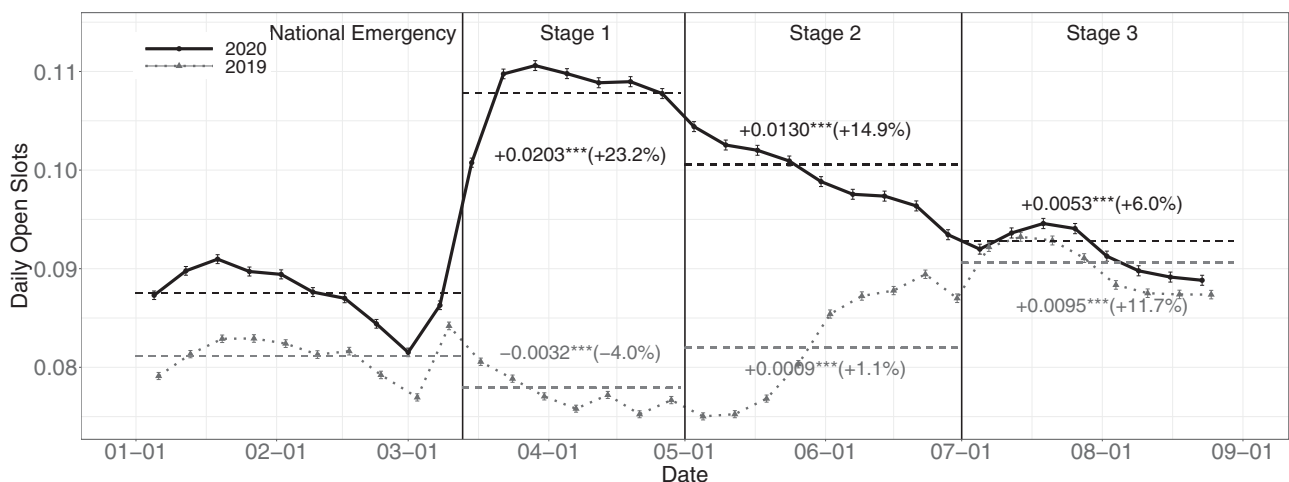
rate (equal to the cumulative number of deaths/cumulative number of cases). The original data of these variables are at the state-day level. Considering the fact that these variables have large fluctuations and strong seasonality within a week, we use their weekly mean in our analysis. We also obtain state-month-level unemployment rates from the U.S. Bureau of Labor Economics. Table A2(a) in the online appendix provides the definitions and sources of these variables, and Table A2(b) in the online appendix provides their summary statistics.

To analyze the impact of NPIs, we obtain the starting and ending dates of nonpharmaceutical policy interventions at the state level from the data set maintained by Raifman et al. (2020).¹² Among the large number of nonpharmaceutical policy interventions, we select the ones that are related to the labor market and may affect gig workers’ labor supply on the platform, including SIP, closure of K–12 schools, closure of day cares, closure of nonessential businesses, and closure of restaurants. These policies all restrict certain groups of people from working on-site. Furthermore, closure of K–12 schools and closure of day cares make kids stay at home. Then, their parents may need to care for them at home and take gig jobs for its flexibility. Thus, the implementations of these NPIs may increase the labor supply to the gig economy. The detailed descriptions of these NPIs and the corresponding variables are provided in Table A2(c) in the online appendix.

3. Impact of COVID-19 on (the Quantity of) Labor Supply

In this section, we investigate how online gig workers change their (quantity of) labor supply during the

Figure 1. How Gig Workers’ Labor Supply Changes during the Pandemic



Notes. The figure shows the weekly average value of the outcome variable because there is an obvious seasonality pattern within each week. The error bars represent the standard error of the weekly average value. The dashed horizontal lines represent the average value of the outcome variable in the period. The outcome variable *Daily Open Slots* has been normalized, so we should mainly focus on the relative change here.

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Table 1. How Labor Supply per Worker Changes During the Pandemic

	Outcome variable: <i>Daily Open Slots</i>			
	(1)	(2)	(3)	(4)
<i>Stage 1</i>	0.0203*** (0.0005)	0.0167*** (0.0005)	0.0165*** (0.0005)	
<i>Stage 2</i>	0.0130*** (0.0005)	0.0051*** (0.0004)	0.0050*** (0.0004)	
<i>Stage 3</i>	0.0053*** (0.0006)	−0.0088*** (0.0005)	−0.0092*** (0.0005)	
<i>Stage 1</i> × <i>Year2020</i>				0.0230*** (0.0006)
<i>Stage 2</i> × <i>Year2020</i>				0.0097*** (0.0006)
<i>Stage 3</i> × <i>Year2020</i>				−0.0075*** (0.0006)
<i>Year2020</i>				−0.0118*** (0.0006)
Constant	0.0875*** (0.0009)			
Worker FE		Yes	Yes	Yes
Day of week FE			Yes	
Day of year FE				Yes
Observations	8,165,521	8,165,521	8,165,521	16,597,901
Adjusted R^2	0.0058	0.5765	0.5809	0.5261

Notes. All analyses are performed at the worker-day level. Columns (1)–(3) use the data of 2020. Column (4) is a year to year DiD regression using the data of both 2019 and 2020. FE stands for “fixed effects”. Standard errors are clustered at the state level and are shown in parentheses. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

pandemic. Following the literature of the COVID-19 pandemic (e.g., Simonov et al. 2020, Ananyev et al. 2021, Bazzi et al. 2021), we take March 13, 2020, the date on which the national emergency was declared, as the major signal of the outbreak of COVID-19 in the United States.

Figure 1 plots how the average labor supply of gig workers on the platform, measured by the average number of slots opened by a worker per day, changes during January to August 2020 and in the same period of time in 2019. It shows that in 2020, the average labor supply per worker is relatively stable before the declaration of the national emergency, but it increases sharply after the national emergency and keeps at the high level in late March and April; the average labor supply per worker gradually declines in May and June and stays at a relatively lower level in July and August. Compared with 2019, the average labor supply per worker in 2020 is slightly higher but has a similar trend to that of 2019 before the date of national emergency; the gap in labor supply between 2020 and 2019 becomes significantly larger after the national emergency and keeps so until the end of April. The gap then gradually shrinks in May and June and becomes very small in July and August. Guided by the observed pattern, we define the period from March 13

to the end of April as stage 1 after the outbreak of the pandemic, May and June as stage 2, and July and August as stage 3.

To formally quantify the change in labor supply in each stage after the outbreak of the pandemic, we conduct the following regression analysis utilizing worker-day level data during January to August 2020 (as shown in column (1) of Table 1):

$$\text{Daily Open Slots}_{it} = \beta_0 + \beta_1 \text{Stage } 1_t + \beta_2 \text{Stage } 2_t + \beta_3 \text{Stage } 3_t + \epsilon_{it}. \quad (1)$$

The outcome variable *Daily Open Slots* means the number of slots opened by a worker in a day and the indicators $\text{Stage } j_t$ represent whether day t is in stage j ($j = 1, 2, 3$). The constant term β_0 represents the average labor supply before the pandemic (i.e., before March 13) in 2020, and the coefficient β_j represents the difference between the average labor supply per worker in stage j versus that of before the pandemic ($j = 1, 2, 3$). The regression result (column (1) of Table 1) shows that compared with before the pandemic, the average labor supply per worker in stage 1 is 23.2% higher, the average labor supply per worker in stage 2 is 14.9% higher, and the average labor supply per worker in stage 3 is only 6.0% higher (absolute and

relative changes are shown in Figure 1). If we include worker fixed effects to control for worker heterogeneity and include day of week fixed effects to control for weekly seasonality (columns (2) and (3)), the observation still holds, but the coefficient of “stage 3” becomes negative, meaning that workers actually decrease their labor supply in July and August if worker heterogeneity is taken into consideration. In Table A3 in the online appendix, we run the same set of regressions using month dummies as the explanatory variables. The coefficients of “Month 3” and “Month 4” are significantly positive and the largest,¹³ the coefficients of “Month 5” and “Month 6” are still significantly positive but become smaller, and the coefficients of “Month 7” and “Month 8” become negative if worker fixed effects are controlled for. The monthly regression results show consistent patterns with those given by Table 1, confirming our definition of stages. In Table A4 in the online appendix, we also conduct placebo tests in which we change the cutoff dates between stages to 5 days earlier/later or 10 days earlier/later, and our main findings do not change qualitatively.

To tease out any possible seasonality effects that may drive the change in gig workers’ labor supply, we use the same period of time in 2019 as the control group and compare it with 2020 following the idea of year to year DiD analysis (Bandiera et al. 2005). We run the following DiD regression to quantify the change in gig workers’ labor supply in each month during the pandemic after seasonality has been controlled for (column (4) of Table 1):

$$\begin{aligned} \text{Daily Open Slots}_{it} = & \beta_0 \text{Year2020}_t + \beta_1 \text{Stage } 1_t \\ & \times \text{Year2020}_t + \beta_2 \text{Stage } 2_t \\ & \times \text{Year2020}_t + \beta_3 \text{Stage } 3_t \\ & \times \text{Year2020}_t + \tau_t + s_i + \epsilon_{it}. \end{aligned} \quad (2)$$

Year2020_t is a dummy variable that equals one if the observation is in the year of 2020. $\text{Stage } j_t$ is a dummy variable that equals one if the date is in stage j in 2020 or in the same range of days in 2019. The terms τ_t and s_i denote the day of year fixed effects and worker fixed effects, respectively. Notice that the stage dummies are absorbed by the day of year fixed effect. The coefficient of Year2020 (β_0) captures the difference in average labor supply between stage 0 (i.e., days before March 13) of 2020 and stage 0 of 2019. The coefficients of the interactions between Year2020 and three stage dummies ($\beta_j, j = 1, 2, 3$) reflect the change in gig workers’ average labor supply in the three stages compared with before the pandemic in 2020, after the seasonality (changes that also happened in 2019) and worker heterogeneity have been controlled for. The estimated impacts are close to the previous estimates in columns (2) and (3), indicating that the change in gig workers’ labor supply after the outbreak of the pandemic is not driven by seasonality.

We further decompose whether the increase in gig workers’ labor supply is because they have more days available to work on the platform or because they work more intensively on their available days. We define that a gig worker is available on a day if she opens at least one slot that day (denoted by a dummy variable *Daily Availability*) and define working intensity as the number of slots opened by a gig worker in an available day (denoted by *Working Intensity*). Table A5 in the online appendix quantifies how *Daily Availability* and *Working Intensity* change during the pandemic. We can see that workers’ availability increases by 10.2%, 7.2%, and 3.9% in stage 1, stage 2, and stage 3, respectively, and the average working intensity increases by 11.7%, 7.2%, and 2.1% in the three stages, respectively. Thus, these two factors each contribute about half of the increase in the labor supply.

To summarize, the average labor supply of gig workers on the platform increases sharply after the outbreak of the pandemic and stays at the high level until the end of April; the increase becomes smaller in May and June, and the increase disappears in July and August after controlling for worker heterogeneity and seasonality. The increase in workers’ availability and their working intensity each contribute about half of the increase in labor supply.

4. Possible Mechanisms

In the previous section, we find that gig workers on the platform sharply increased their quantity of labor supply after the outbreak of the pandemic. The increase lasts from March 13, 2020 to June 2020 (stage 1 and stage 2) but disappeared after July 2020 (stage 3). In this section, we investigate the possible underlying mechanisms that drive the increase in gig workers’ labor supply in stage 1 and stage 2.

We first analyze whether the increase in gig workers’ labor supply is in response to certain changes that happen to the platform after the outbreak of the pandemic. In particular, the pandemic may affect the demand side of the platform or the entry of new workers to the platform, which can in turn influence the labor supply decision of existing workers. The detailed analysis is available in Online Appendix A1. In short, we first graphically show that the increase in demand has a much smaller magnitude than the increase in existing workers’ labor supply, and we further conduct a mediation analysis to show that the change in demand cannot explain the increase in gig workers’ labor supply. We also conduct mediation analyses to show that the increase in existing gig workers’ labor supply cannot be explained by the entry of new workers.

4.1. Three Factors Brought by the Pandemic

As explained, the increase in gig workers’ labor supply was not because of a higher demand or excessive entry of new workers brought by the pandemic.

Because there were no other major changes that happened to the platform right after the outbreak of the pandemic, the increase in gig workers' labor supply to the platform is unlikely to be driven by other changes to the platform. Instead, it is likely to be a direct response to the pandemic itself and certain consequences brought by the pandemic. In particular, there are three factors brought by the pandemic that can lead to the increase in gig workers' labor supply after the outbreak of the pandemic.

The first factor is the risk of contracting COVID-19. Many gig workers have other full-time or part-time jobs besides working on the gig economy platform. Many people were concerned about the exposure to coronavirus when working on-site.¹⁴ Thus, when the pandemic is more severe, gig workers may choose to switch from offline jobs to online jobs to reduce the risk of contracting COVID-19. That is, gig workers' labor supply on the online gig platform may increase when the pandemic is more severe.

The second factor is unemployment. The unemployment rate rises after the outbreak of the pandemic.¹⁵ In April 2020, the unemployment rate hit 14.8%—the highest record since 1948.¹⁶ When gig workers lose their other jobs, they may work more on the gig economy platform because they have more time to work online and also because they need to remedy their loss of income (Huang et al. 2020).

The third factor is the NPIs. There have been numerous NPIs implemented across the country aiming to reduce the spread of COVID-19 (Raifman et al. 2020), including stay-at-home orders, closures of schools, closures of nonessential businesses, etc. Some NPIs make people unable to work on-site (for example, stay-at-home orders and/or closures of restaurants make waiters and waitresses unable to work on-site), and these people may turn to working for the online gig economy platform. Some NPIs make people have to leave their traditional 9–5 jobs to care for their family (for example, closures of day cares and K–12 schools make parents have to take care of their kids at home), and they may take gig jobs, which have a flexible working schedule.¹⁷

Now, we investigate whether and how much these three factors can explain the increase in gig workers' labor supply to the platform. For the first factor, we choose six variables to represent the severity of the COVID-19 pandemic, which can measure the risk of contracting COVID-19. As mentioned in Section 2, considering the strong weekly seasonality of these measures, we use their weekly average in our regression analyses. The second factor is naturally reflected by the unemployment rate at the state level. For the third factor, among the numerous NPIs implemented throughout the country, we choose five NPIs that are directly related to the labor market and are most likely

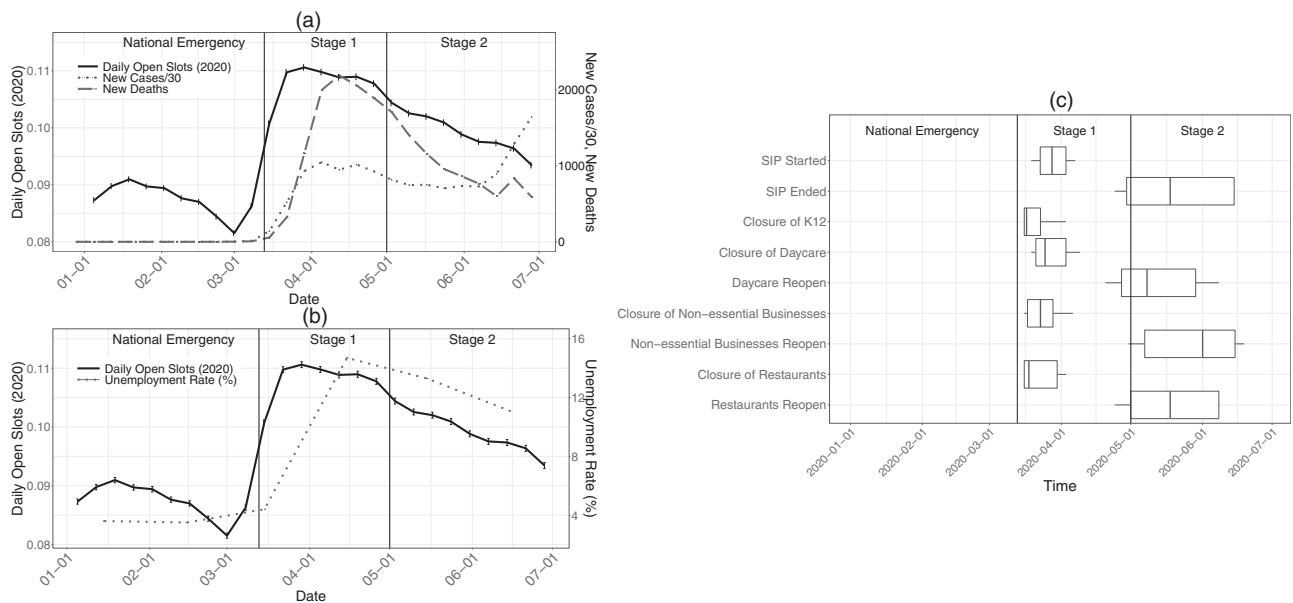
to affect the labor supply of gig workers, and we collect the dates on which these NPIs started and ended. (See Section 2.2 for the detailed illustrations of how we select the COVID-19 severity measures and NPIs. Their definitions and sources are provided in Table A2 in the online appendix.) All these variables are at the state level. We notice that COVID-19 spreads to different states at different paces, and different states also have different magnitudes of increases in the unemployment rate and different starting and ending dates of the selected NPIs. These variations across states allow us to causally identify the impact of each factor on gig workers' change in labor supply using state-level panel analyses with worker fixed effects, and then, we can quantify how much each of these factors can explain gig workers' change in labor supply.

We first graphically compare the average labor supply of gig workers on the platform with the trend of the severity of the COVID-19 pandemic¹⁸ (Figure 2(a)) as well as the trend of unemployment rate (Figure 2(b)). We can see that the trend of the average labor supply is very close to the trend of new deaths in stage 1 and stage 2. The difference between the trend of average labor supply and the trend of new cases is larger, especially toward the end of stage 2 when the number of new cases soared to a new height but the average labor supply kept decreasing. In contrast, the trend of average labor supply is very similar to the trend of unemployment rate. Figure 2(c) shows the range of dates when the related NPIs take effect. We can see that these related NPIs all start in stage 1. The majority of states end these policies (except for "closure of K–12," which does not end in our data range) in stage 2. This pattern is consistent with the fact that the average labor supply sharply increases in stage 1 and gradually decreases in stage 2. The graphical comparisons indicate that unemployment seems to better explain the change in average labor supply than the pandemic severity does. Within the two measures of the COVID-19 pandemic, *New Deaths* rather than *New Cases* explains the change in average labor supply better. The implementation of NPIs may also explain the increase of average labor supply in stage 1 and its gradual decline in stage 2.

The graphical comparisons can only give us a rough idea of whether each factor may explain the increase in average labor supply. To better identify whether and how much each factor explains the increase in gig workers' labor supply, we conduct mediation analyses for the three factors, as reported in Tables 2–4.

The first step of mediation analysis is to show the relation between the dependent variable (*Daily Open Slots*) and the independent variables (the stage dummies), which is reported in column (1) of Table 3. It shows that, after controlling for weekly seasonality and worker heterogeneity, gig workers' labor supply

Figure 2. Severity of the COVID-19 Pandemic, Unemployment, and NPIs



Notes. In Figure (a), *New Cases* and *New Deaths* mean the daily numbers of new cases and new deaths in the United States, respectively. For *New Cases*, *New Deaths*, and *Daily Open Slots*, we show the weekly average value of the variables in the figures because the original daily values have an obvious seasonality pattern within each week. The error bars represent the standard error of the weekly average value of *Daily Open Slots*. The *Unemployment Rate* in Figure (b) represents the monthly national unemployment rate. In Figure (c), the boxes and whiskers show the range of time that each NPI takes effect across states. The left side of each box is the 10th percentile of the starting time of the NPI across states, and the right side is the 90th percentile. The middle vertical line shows the median of the starting time of the NPI across states. The two ends of the whiskers show the minimum and maximum of the starting times of the NPI. Notice that *SIP Started*, *Closure of K-12*, *Closure of Day Care*, *Closure of Nonessential Business*, and *Closure of Restaurants* indicate the start of policies and that *SIP Ended*, *Day Care Reopen*, *Nonessential Businesses Reopen*, and *Restaurants Reopen* indicate the end of policies. Within our data observation period, no states have reopened K-12 schools, so there is no “K-12 Reopen” here.

on the platform increases in stage 1 and stage 2, and the increase is larger in stage 1.

The second step of mediation analysis is to show the relation between the mediators (the six measures of the COVID-19 pandemic, the unemployment rate, and the NPI dummies) and the independent variables. We regress the six measures of COVID-19 severity as well as the unemployment rate on the stage dummies, as

reported in Table 2. As expected, both stage dummies have significantly positive coefficients in all seven regressions, meaning that these seven measures all significantly increased after the outbreak of the COVID-19 pandemic. We cannot do the regression analyses to establish the relation between the NPI dummies and stage dummies, but we have shown in Figure 2(c) that all the related NPIs were started in stage 1 and ended

Table 2. Mediation Analysis: The Second Step

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	<i>New Cases per 1K</i>	<i>New Deaths per 1K</i>	<i>Test Positive Rate</i>	<i>COVID-19 Inpatients per 1K</i>	<i>COVID-19 Inpatient Beds Utilization</i>	<i>Mortality Rate</i>	<i>Unemployment</i>
<i>Stage 1</i>	0.0554*** (0.0086)	0.0028*** (0.0006)	0.1176*** (0.0109)	0.0799*** (0.0111)	0.0640*** (0.0081)	0.0245*** (0.0015)	6.5881*** (0.3591)
<i>Stage 2</i>	0.0709*** (0.0051)	0.0029*** (0.0004)	0.0577*** (0.0044)	0.1268*** (0.0131)	0.0599*** (0.0062)	0.0408*** (0.0028)	7.3425*** (0.4692)
Constant	0.0030 (0.0035)	−0.0007* (0.0003)	0.0311*** (0.0037)	−0.0079 (0.0070)	−0.0039 (0.0039)	−0.0034** (0.0012)	2.2576*** (0.2507)
State FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	9,256	9,256	9,256	9,256	9,256	9,256	9,195
Adjusted R ²	0.4195	0.3815	0.4956	0.4872	0.4608	0.6664	0.5745

Notes. The regression analyses are performed at the state-day level. For the six measures of pandemic severity (columns (1)–(6)), we use the mean value of current week. Column (7) has fewer observations because the unemployment rates of Puerto Rico are missing for March and April 2020. FE stands for “fixed effects”. Standard errors are clustered at the state level and are shown in parentheses.

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

in stage 2 (with a few exceptional states that ended SIP, reopened day cares, or reopened restaurants toward the end of stage 1).

The third step of mediation analysis is to regress the dependent variable on the mediators as well as the independent variables. A valid mediator should satisfy that (1) its coefficient is significant in this regression and that (2) after adding this mediator, the

coefficients of the independent variables become smaller compared with in the first step. Notice that these three factors are interrelated. The severity of the COVID-19 pandemic may not only make gig workers increase their labor supply on the platform directly, but also have an indirect impact by influencing governments' NPI decisions and influencing the unemployment rate. The NPIs can have an impact on the

Table 3. Mediation Analysis: The First and Third Steps

	Outcome variable: <i>Daily Open Slots</i>							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Stage 1</i>	0.0165*** (0.0005)	0.0028*** (0.0007)	0.0133*** (0.0008)	0.0036*** (0.0007)	0.0138*** (0.0007)	0.0028*** (0.0007)	0.0035*** (0.0008)	0.0121*** (0.0008)
<i>Stage 2</i>	0.0049*** (0.0004)	−0.0043*** (0.0010)	0.0039*** (0.0010)	−0.0051*** (0.0010)	0.0019*** (0.0005)	−0.0042*** (0.0010)	−0.0051*** (0.0010)	0.0032*** (0.0008)
<i>New Cases per 1K</i>		−0.0432*** (0.0097)	−0.0445*** (0.0118)			−0.0427*** (0.0100)		−0.0430*** (0.0100)
<i>New Deaths per 1K</i>		0.3556** (0.1026)	0.2873 (0.1529)			0.3547** (0.1017)		0.3336** (0.1240)
<i>Test Positive Rate</i>		0.0107** (0.0038)	0.0204*** (0.0054)			0.0106** (0.0039)		0.0183*** (0.0050)
<i>COVID-19 Inpatients per 1K</i>		−0.0064 (0.0055)	−0.0018 (0.0065)			−0.0069 (0.0056)		−0.0072 (0.0062)
<i>COVID-19 Inpatient Beds Utilization</i>		0.0265*** (0.0072)	0.0355 (0.0178)			0.0262*** (0.0071)		0.0315* (0.0154)
<i>Mortality Rate</i>		−0.0529*** (0.0131)	−0.0172 (0.0252)			−0.0546*** (0.0133)		−0.0440** (0.0158)
<i>Unemployment</i>		−0.0000 (0.0001)		−0.0000 (0.0001)	0.0004*** (0.0000)			0.0004*** (0.0001)
<i>SIP Started</i>		0.0003 (0.0006)		0.0001 (0.0007)		0.0002 (0.0007)	−0.0000 (0.0007)	
<i>SIP Ended</i>		−0.0003 (0.0013)		−0.0004 (0.0014)		−0.0002 (0.0014)	−0.0004 (0.0015)	
<i>Closure of K–12</i>		0.0118*** (0.0011)		0.0115*** (0.0014)		0.0118*** (0.0011)	0.0114*** (0.0014)	
<i>Closure of Day Care</i>		0.0026** (0.0008)		0.0028*** (0.0007)		0.0025** (0.0008)	0.0027*** (0.0007)	
<i>Day Care Reopen</i>		−0.0029*** (0.0008)		−0.0033*** (0.0008)		−0.0029*** (0.0008)	−0.0033*** (0.0008)	
<i>Closure of Nonessential Businesses</i>		0.0013 (0.0009)		0.0003 (0.0011)		0.0010 (0.0007)	0.0001 (0.0007)	
<i>Nonessential Businesses Reopen</i>		−0.0021* (0.0009)		−0.0027** (0.0009)		−0.0021* (0.0009)	−0.0027** (0.0009)	
<i>Closure of Restaurants</i>		0.0019 (0.0012)		0.0029* (0.0012)		0.0020 (0.0012)	0.0029* (0.0012)	
<i>Restaurants Reopen</i>		−0.0026 (0.0014)		−0.0030 (0.0015)		−0.0026 (0.0014)	−0.0030* (0.0015)	
Day of week FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Worker FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	6,437,768	6,157,707	6,157,768	6,437,707	6,437,707	6,157,768	6,437,768	6,157,707
Adjusted R ²	0.5976	0.6032	0.6025	0.5984	0.5977	0.6032	0.5984	0.6026

Notes. The regression analyses are performed at the worker-day level. For the six measures of pandemic severity (columns (1)–(6)), we use the mean value of the previous week, and this is also why columns (2), (3), (6), and (8) have fewer observations. The unemployment rates of Puerto Rico are missing for March and April 2020, and therefore, the columns involving *Unemployment* (columns (2), (4), and (5)) have fewer observations. FE stands for “fixed effects”. Standard errors are clustered at the state level and are shown in parentheses.

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

unemployment rate and then, in turn, affect gig workers' labor supply. On the other hand, the change in the unemployment rate may also have an impact on governments' NPI decisions and then, influence the labor supply on the platform. Figure A2 in the online appendix plots the possible relationships between the three factors and gig workers' labor supply on the platform. For each mediator, if we only add this mediator to the regression and compare it with column (1) of Table 3, the estimated mediation effect is the total mediation effect of this mediator. That is, it captures the direct effect of this mediator on gig workers' labor supply plus the indirect effect of this mediator on the labor supply through influencing the other two mediators. If we want to estimate the direct mediation effect of this mediator, we need to control for the other two mediators in the regression (VanderWeele and Vansteelandt 2014, Vansteelandt and Daniel 2017). That is, we first add the other two mediators to the regression and then, further include this focal mediator in the regression. The decrease in the coefficients of stage 1 and stage 2 between these two regressions reflects the direct mediation effect of this focal mediator on gig workers' labor supply after the other two factors have been controlled for.¹⁹

Considering the reality that it may take some time for people to perceive the changes in the three factors, we test if it works better to use the values of three factors in the current period or in the previous week in the mediation analyses. In Table A7 in the online appendix, we compare how much each of the three factors can explain the increase in gig workers' labor supply if their values in the current period (panel A) or in a week ago (panel B) are used in the mediation analyses. We find that for the six measures of pandemic severity, their values in the previous week can better explain the increase in gig workers' labor supply (i.e., have a larger mediation effect). For unemployment and NPIs, their values in the current period work better.²⁰ Therefore, we use the pandemic severity of the previous week and the current values of unemployment and NPIs in our mediation analyses. In Table A7(c) in the online appendix, we show that adding both the current values and the previous

values does not work better than our choice of combinations (as given in Tables 3–4).

In column (3) of Table 3, we add the six measures of the COVID-19 pandemic severity (of the previous week) to the regression. Compared with column (1), stage 1's coefficient is reduced by 19.4%, and stage 2's coefficient gets reduced by 20.4% (calculated in Table 4)—these are the total mediation effects of pandemic severity in the two stages. Column (4) of Table 4 reports the direct mediation effect of pandemic severity, which comes from the comparison in stage 1 and stage 2's coefficients between column (4) and column (2) in Table 3. We can see that pandemic severity has a very small direct mediation effect in stage 1, and its direct effect in stage 2 is even negative. These results indicate that the direct risk of contracting COVID-19 plays a very small role in the increase in gig workers' labor supply on the online platform. Instead, the pandemic severity affects gig workers' labor supply through influencing the NPI policies and unemployment.

Notice that the coefficients of the six measures of pandemic severity in column (3) of Table 3 may not be meaningful because of colinearity. In columns (2)–(7) of Table A8 in the online appendix, we show that when each of the six measures of pandemic severity is added to regression separately, all six variables have positive coefficients, among which *New Deaths per 1K*, *Test Positive Rate*, *COVID-19 Inpatients per 1K*, and *COVID-19 Inpatient Beds Utilization* have significantly positive coefficients. The results indicate that when considering the risk of getting COVID-19, people care about how likely they are to get COVID-19 and the consequence of the disease (i.e., deaths) as well as how likely they can get inpatient care if they get severely ill.

Column (5) of Tables 3 and 4 estimates the total mediation effect of unemployment. The unemployment rate has a significantly positive coefficient. Adding the unemployment rate reduces stage 1's coefficient by 16.4% and reduces stage 2's coefficient by 61.2%. Column (6) estimates the direct mediation effect of unemployment. We can see that the direct effect of unemployment is almost negligible. This does not mean that unemployment has no impact on gig workers' labor supply on the platform; otherwise, its total mediation effect cannot be that

Table 4. Mediation Analysis: Calculation of Total and Direct Mediation Effects

	Total effects (1)	Explained by three mediators (2)	Explained by severity		Explained by unemployment		Explained by policies	
			Total (3)	Direct (4)	Total (5)	Direct (6)	Total (7)	Direct (8)
Stage 1	0.0165	0.0137 83.0%	0.0032 19.4%	0.0008 4.8%	0.0027 16.4%	0.0000 0.0%	0.0130 78.8%	0.0093 56.4%
Stage 2	0.0049	0.0092 187.8%	0.0010 20.4%	−0.0008 −16.3%	0.0030 61.2%	0.0001 2.0%	0.0100 204.1%	0.0075 153.1%

Notes. Based on the regression results in Table 3, this table calculates how much the increase in gig workers' labor supply in stage 1 and stage 2 (column (1)) can be explained by the total or direct effect of each mediator (columns (3)–(8)) or the three mediators as a whole (column (2)), with the percentage value below the absolute value. FE stands for “fixed effects”.

significant. It should be because that most of the change in the unemployment rate in stage 1 and stage 2 is driven by the NPIs,²¹ and therefore, when controlling for NPIs, the direct channel of “stage 1/2 → unemployment → labor supply” is negligible.

Columns (7) and (8) of Tables 3 and 4 estimate the total mediation effect and direct mediation effect of the selected NPIs, respectively. Column (7) shows that adding these NPI dummies reduces stage 1's coefficient by 78.8%, and drives stage 2's coefficient to negative. The direct effect of NPIs (as shown in column (8)) is close to the total effect of NPIs. These results indicate that the selected NPI policies can largely explain the increase in gig workers' labor supply on the platform. If there were not these NPI dummies, the labor supply of existing gig workers in stage 2 would have been lower compared with before the pandemic. Furthermore, most of the impact of NPIs on gig workers' labor supply is direct impact rather than through influencing unemployment.

In columns (8)–(12) of Table A8 in the online appendix, we add the dummies of these NPIs to the regression separately. All the dummies of starting a policy (*SIP Started*, *Closure of K–12*, *Closure of Nonessential of Business*, and *Closure of Restaurants*) have a significantly positive coefficient, and all the dummies of ending a policy (*SIP Ended*, *Day Care Reopen*, *Nonessential Businesses Reopen*, and *Restaurants Reopen*) have a significantly negative coefficient. These results are consistent with our intuition: when schools or businesses were closed, gig workers increase their labor supply on the online platform, and when schools or businesses were reopened, the labor supply decreased accordingly. Columns (10)–(14) of Table A9 in the online appendix estimate the direct effect of each of the NPIs, showing that among these NPI policies, closure of K–12 explains most of the increase in gig workers' labor supply.

Lastly, the comparison between column (1) and column (2) of Table 3 allows us to estimate how much of the increase in labor supply in stage 1 and stage 2 can be explained by the three factors as a whole. We can see that the three factors explain 83.0% of the increase in stage 1 and 187.8% of the increase in stage 2. In other words, if there were no such three factors brought by the pandemic, gig workers' labor supply to the platform would still increase but with a much smaller magnitude in stage 1, and it would decrease significantly in stage 2 (compared with before the pandemic). The size of the total mediation effect of the three factors is only slightly larger than the mediation effect of NPI policies (column (7)). This finding indicates that the NPI policies can explain most of the increase in gig workers' labor supply brought by the pandemic.

To summarize, our analyses indicate that the increase in gig workers' labor supply on the platform after the outbreak of the pandemic was not driven by the

demand side or excessive entry of new workers. There are three factors brought by the pandemic that may explain the increase in gig workers' labor supply: the risk of contracting COVID-19, the rise of unemployment, and the implementation of NPI policies aiming to slow down the spread of the disease. Our mediation analyses show that among the three factors, NPI policies explain most of the increase in gig workers' labor supply. Gig workers increase their labor supply on the online platform not because of the risk of contracting COVID-19. Instead, the pandemic severity affects gig workers' labor supply through influencing the NPI policies. The impact of unemployment is much smaller than the impact of NPI policies, indicating that the increase in gig workers' labor supply is more driven by temporary changes in working arrangements because of the policies rather than relatively long-term changes in employment situations. This is consistent with the fact that gig workers' labor supply quickly gets lower in stage 3, when all these work-related NPIs have ended.

5. Earning Potential and Change in Labor Supply

In this section, we will investigate the relationship between gig workers' earning potential on the platform and their changes in labor supply during the pandemic. By earning potential of a gig worker, we do not mean how much money she actually earns on the platform but how easy/difficult it is for her to earn money on the platform. A worker's earning potential on the platform is determined by two factors. One is the hourly wage the worker earns on the platform. When a worker enters the platform, the platform sets an hourly wage for her based on her performance at the interview. Once every year, the platform will examine the worker's performance over the past year and decides whether to increase her hourly wage by \$1.²² The other factor that determines the earning potential is the booking rate (i.e., the probability of the worker's opened slot being booked by a student).

We first examine the relationship between gig workers' hourly wage and their change in labor supply. We regress *Daily Open Slots* on the dummies of three stages and hourly wage of the worker as well as the interaction terms between the stage dummies and hourly wage (column (1) of Table 5); the result shows that high-hourly wage workers have a smaller increase in labor supply, especially in stage 1 and stage 2.²³ However, this result might be driven by workers' preexisting differences—for example, workers' with lower pre-COVID-19 labor supply may have a larger space for increase in labor supply. To control for workers' preexisting differences, we add worker fixed effects into the regression (column (2) of Table 5). Now, the

Table 5. Relationship between Earning Potential and Change in Labor Supply

	Outcome variable: <i>Daily Open Slots</i>			
	(1)	(2)	(3)	(4)
<i>Stage 1</i>	0.0204*** (0.0005)	0.0169*** (0.0005)	0.0192*** (0.0005)	0.0157*** (0.0005)
<i>Stage 2</i>	0.0131*** (0.0005)	0.0053*** (0.0004)	0.0104*** (0.0005)	0.0032*** (0.0005)
<i>Stage 3</i>	0.0045*** (0.0006)	−0.0091*** (0.0005)	0.0011 (0.0006)	−0.0120*** (0.0005)
<i>Stage 1</i> × <i>Hourly Pay</i>	−0.0011*** (0.0002)	−0.0004 (0.0002)		
<i>Stage 2</i> × <i>Hourly Pay</i>	−0.0008* (0.0004)	0.0011*** (0.0003)		
<i>Stage 3</i> × <i>Hourly Pay</i>	−0.0007 (0.0004)	0.0026*** (0.0003)		
<i>Stage 1</i> × <i>Pre-COVID-19 Booking Rate</i>			−0.0035*** (0.0005)	0.0001 (0.0005)
<i>Stage 2</i> × <i>Pre-COVID-19 Booking Rate</i>			−0.0024*** (0.0005)	0.0047*** (0.0005)
<i>Stage 3</i> × <i>Pre-COVID-19 Booking Rate</i>			−0.0022*** (0.0005)	0.0092*** (0.0006)
<i>Hourly Pay</i>	0.0113*** (0.0006)			
<i>Pre-COVID-19 Booking Rate</i>			0.0019** (0.0006)	
Day of week FE	Yes	Yes	Yes	Yes
Worker FE		Yes		Yes
Observations	8,094,036	8,094,036	7,785,092	7,785,092
Adjusted R^2	0.0212	0.5822	0.0107	0.5781

Notes. All analyses are performed at the worker-day level using the data of 2020. *Hourly Pay* and *Pre-COVID-19 Booking Rate* have been normalized to the standard normal distribution. The hourly pay information is missing for 613 workers; 2,113 workers have no *Pre-COVID-19* opened slots and therefore, have missing *Pre-COVID-19* booking rates. Therefore, the numbers of observations are smaller than in columns (1)–(3) of Table 1. FE stands for “fixed effects”. Standard errors are clustered at the state level and are shown in parentheses.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

coefficients of the three stage dummies become positive, positive, and negative (i.e., in the same directions as in Table 1, column (2), where worker fixed effects have been included). The coefficient of *stage 1* × *Hourly Wage* is statistically insignificant. The coefficients of *stage 2* × *Hourly Wage* and *stage 3* × *Hourly Wage* are both significantly positive, and the latter one has a smaller magnitude than the coefficient of stage 3. This result indicates that after controlling for preexisting worker heterogeneity (i.e., for workers with the same pre-COVID-19 labor supply), the hourly wage does not affect workers’ change in labor supply in stage 1; in stage 2, workers with higher hourly wages increase their labor supply more, and in stage 3, workers with higher hourly wages are less likely to reduce their labor supply.

Regarding the second factor, the booking rate, we assume that a worker’s booking rate before the pandemic forms her belief about how likely her opened slots can be booked by students.²⁴ We calculate the

pre-COVID-19 booking rate of each worker using the worker’s total number of booked slots divided by her total number of opened slots before the pandemic and perform similar analyses as for hourly wage. The regression model without worker fixed effects shows that workers with high/low pre-COVID-19 booking rates have similar pre-COVID-19 average labor supply, but workers with low pre-COVID-19 booking rates have a larger increase in labor supply after the pandemic (column (3) of Table 5).²⁵ However, when we include worker fixed effects in the regression (column (4) of Table 5), the signs of the interaction terms get reversed and become consistent with column (2). That is, for workers with the same pre-COVID-19 average labor supply, in stage 1, workers open more slots regardless of the perceived booking rate; in stage 2, workers with a higher perceived booking rate increase their labor supply more, and in stage 3, workers with a higher perceived booking rate are less likely to reduce their labor supply.

To summarize, after controlling for workers' preexisting differences, we find that earning potential does not affect the change in labor supply in the early stage (stage 1) of the pandemic, but workers with higher earning potential are more likely to increase or less likely to decrease their labor supply in the later stages (stage 2 and stage 3). That is, in the early stage of the pandemic, gig workers sharply increase their labor supply regardless of their earning potential. In the later stages (i.e., stage 2 and stage 3), the pandemic is still severe, but as the work-related NPIs end and the unemployment rate declines, workers have more alternative working opportunities; therefore, workers with a lower earning potential are more likely to reduce their labor supply on the platform as more outside working opportunities become available. This is consistent with our finding in Section 4 that working opportunities and working arrangements largely explain gig workers' change in labor supply.

6. Quality of Labor Supply

In this section, we investigate how the quality of labor supply changes after the outbreak of the pandemic.

There are two major ways to measure a gig worker's quality of labor supply on the platform. The first one is whether the worker fails to fulfill a booked slot because of her own reasons. A worker may fail to fulfill a booked slot if she cancels the class, does not show up during the class, or has a technology issue because of her own reasons. We define a binary variable "*Worker Problem*," which equals one if a worker has one of these issues and fails to fulfill a booked slot as a result.²⁶ The second one is the student's rating of the class after the class is finished (*Class Rating*). The class ratings are integers ranging from one to five, and a higher rating means the student is more satisfied

with the class. Because the majority of the ratings are five, a rating lower than five usually indicates that the student is very unsatisfied with the class. Thus, we define a binary variable *Low Rating*, which equals one if the rating is lower than five. If a worker increases her quality of work, then she should be less likely to have a worker problem, should be less likely to have a low rating, and should have a higher average class rating. Table A1(b) in the online appendix provides the summary statistics of the three quality measures.

Our regression analyses are shown in Table 6. All these regressions are based on the class-level data of booked slots of the gig workers in our sample of 2020. We use *Worker Problem*, *Low Rating*, and *Class Rating* as the outcome variables and use the dummies of the three stages as the explanatory variables. Columns (1)–(3) do not include worker fixed effects, showing that compared with before the pandemic, the probability of a worker problem is lower in all three stages, the probability of a low rating is lower in stage 1 and stage 2, and the average class rating is higher in stage 1 and stage 2 (and marginally lower in stage 3). These results indicate that compared with before the pandemic, gig workers' quality of service is generally better during the pandemic, especially in stage 1 and stage 2. However, it is unclear whether it is because workers indeed perform better after the pandemic or because workers with better overall performance are more likely to open time slots. To answer this question, we include worker fixed effects in columns (4)–(6) to control for worker heterogeneity. Now, we see that, compared with before the pandemic, the probability of a worker problem is not significantly different in stage 1 but gets higher in stage 2 and stage 3. Workers are less likely to get a low rating in stage 1 but more likely to get a low rating in stage 3. Workers'

Table 6. How Quality of Work Changes During the Pandemic

	(1) <i>Worker Problem</i>	(2) <i>Low Rating</i>	(3) <i>Class Rating</i>	(4) <i>Worker Problem</i>	(5) <i>Low Rating</i>	(6) <i>Class Rating</i>
<i>Stage 1</i>	−0.0023*** (0.0002)	−0.0011*** (0.0001)	0.0019*** (0.0002)	−0.0001 (0.0001)	−0.0004*** (0.0001)	0.0006*** (0.0001)
<i>Stage 2</i>	−0.0024*** (0.0002)	−0.0011*** (0.0001)	0.0016*** (0.0002)	0.0008*** (0.0002)	−0.0001 (0.0001)	−0.0004* (0.0002)
<i>Stage 3</i>	−0.0015*** (0.0002)	0.0000 (0.0001)	−0.0008* (0.0003)	0.0025*** (0.0002)	0.0010*** (0.0001)	−0.0028*** (0.0003)
Worker FE				Yes	Yes	Yes
Day of week FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	28,523,299	3,268,970	3,268,970	28,522,643	3,267,644	3,267,644
Adjusted R ²	0.0002	0.0002	0.0001	0.0378	0.0226	0.0234

Notes. All analyses are performed at the class level using the data of 2020. There are many missing observations for class ratings (i.e., many students do not submit a rating after a class), and therefore, the number of observations gets lower when *Low Rating* and *Class Rating* are the outcome variables (columns (2), (3), (5), and (6)). The number of observations is slightly smaller in columns (4)–(6) compared with columns (1)–(3) because singleton observations are dropped when worker fixed effects are included. FE stands for "fixed effects". Standard errors are clustered at the state level and are shown in parentheses.

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

average class ratings get higher in stage 1 but lower in stage 2 and stage 3. The results of columns (4)–(6) show that, after controlling for worker heterogeneity, each individual worker on average performs better in stage 1 but performs worse in stage 2 and stage 3.

Now, we investigate whether workers with better performance are more likely to increase their labor supply after the pandemic. We use a worker's average performance before the pandemic shock (i.e., her mean values of *Worker Problem*, *Low Rating*, and *Class Rating* in the pre-COVID-19 time period (before March 13 in 2020)) to measure the worker's intrinsic "quality." Then we regress *Daily Open Slots* on the interaction term between stage dummies and these worker quality measures, as shown in Table A12 in the online appendix. The results show that workers who are less likely to have worker problems before the pandemic increase their labor supply more after the pandemic, especially in stage 2 and more so in stage 3 (column (1)); workers who are less likely to have low ratings (or have higher average ratings) before the pandemic increase their labor supply more in stage 3 (columns (2) and (3)). Overall, the results indicate that high-quality workers increase their labor supply more after the pandemic (or are less likely to lower their labor supply in stage 3).

To summarize, we find that each individual worker tends to perform worse in stage 2 and stage 3 compared with before the pandemic. However, high-quality workers are more likely to increase their labor supply during the pandemic, especially in stage 2 and stage 3. Therefore, the average performance measures get better during the pandemic.

We are also interested in the relationship between quality and quantity of labor supply. That is, when a gig worker opens more slots, does her quality of service tend to be higher or lower? To answer this question, we regress the performance measures on *Daily Open Slots*, controlling for day fixed effects and worker fixed effects (columns (1)–(3) of Table A13 in the online appendix); we further add the interaction terms of stage dummies and *Daily Open Slots* to the regressions to see how the effect differs across different stages of the pandemic (columns (4)–(6) of Table A13 in the online appendix). Including worker fixed effects in the regression allows us to control for worker heterogeneity and focus on the within-worker relationship between quality and quantity. We see that when workers open more slots, they are more likely to have a worker problem (column (1)), and this effect is stronger in stage 1 (column (4)). When workers open more slots, they are also more likely to have a low rating and have lower average class ratings (columns (2) and (3)). This effect is weaker in stage 1 (columns (5) and (6)) but still in the same direction (the absolute value of the coefficient of stage 1 $\times$ *Daily*

Open Slots is smaller than that of *Daily Open Slots*). Overall, we find that for both before and after the pandemic, when a worker provides more labor supply, her quality of labor supply tends to decrease. This effect is smaller in stage 1, which should be because the supply side becomes very competitive in stage 1 as gig workers all suddenly increase their labor supply, and thus, they work hard to maintain the quality of service to attract more students.

In summary, for both before and after the pandemic, when a gig worker provides more labor supply, her quality of labor supply tends to decrease, although this effect is smaller in stage 1 because of severe competition on the supply side. In stage 2 and stage 3, as the NPIs end and the unemployment rate declines, gig workers have more other working options, and the supply side on the platform becomes less competitive; thus, individual gig workers on this platform tend to have worse performance (compared with their own performance before the pandemic). This is consistent with our findings in Sections 4 and 5. However, high-quality workers tend to increase their labor supply more during the pandemic, especially in stage 2 and stage 3. As a result, we see better average performance measures throughout the three stages.

7. Discussion

In this paper, we use data from a large online education platform that has more than 100,000 gig workers in the United States to study the impact of the COVID-19 pandemic on online gig workers' behavior. We quantify the increase in gig workers' labor supply during the pandemic and how long it lasts, and we investigate the key driving forces of the change in their labor supply. We also investigate how they change their quality of labor supply and the relationship between quality and quantity of labor supply as well as the relationship between gig workers' earning potential and their change in labor supply.

We find that the average labor supply of gig workers on the platform increases sharply after the announcement of the national emergency and remains at the high level until the end of April (stage 1); it declines gradually in May and June (stage 2) and stays at a lower level in July and August (stage 3). After we control for worker heterogeneity and seasonality using year to year DiD analyses, the increase in labor supply during stage 1 and stage 2 is still robust, and the increase disappears in stage 3. We also show that the increase in labor supply is not driven by a higher demand or excessive entry of new workers during the pandemic. Further analyses indicate that the risk of contracting COVID-19 plays a negligible role in affecting gig workers' change in labor supply; instead, the loss of working opportunities and change in working

arrangements brought by shocks during the pandemic, especially by nonpharmaceutical interventions, explain most of change in gig workers' labor supply. Gig workers with lower earning potential on the platform have less stable labor supply; although gig workers all increase their labor supply in the early stage of the pandemic, gig workers with lower earning potential are more likely to reduce their labor supply in the later stage when more alternative working opportunities become available. When a gig worker increases her quantity of labor supply, her quality of labor supply decreases. However, high-quality gig workers are more likely to increase their quantity of labor supply during the pandemic, resulting in higher average performance measures during the pandemic compared with before the pandemic. Our findings have important managerial and policy implications, as we will discuss in Section 7.1.

7.1. Managerial and Policy Implications

Our findings have several important managerial and policy implications. First, our findings can help gig economy platforms, especially those operating in an online environment, be prepared for the highly volatile labor supply in a major social disruption. As we have quantified the change in gig workers' labor supply and how long it lasts and also, decomposed how much each driving force contributes to the increase in gig workers' labor supply, our findings can help gig economy platforms better predict the change in gig workers' labor supply when a major social disruption happens in the future, especially for social disruptions that can lead to a rise in unemployment or similar policies, like closures of schools, businesses, etc.

Second, our findings can help gig economy platforms better manage their labor supply. For example, we find that gig workers with a lower earning potential on the platform have a less stable labor supply. Although gig workers all increase their labor supply sharply when the disruption happens, when alternative working opportunities become available, gig workers with lower earning potential are more likely to reduce their labor supply. Thus, if the platform wants to stabilize the labor supply on the platform, the platform should pay more attention to these low-earning potential gig workers. In particular, the platform can help improve their booking rates by giving these workers higher weight in the recommendation algorithm.²⁷ Instead of setting a fixed hourly pay, the platform can also use a dynamic payment scheme for gig workers, which can help stabilize their labor supply on the platform, especially in such major social disruptions.

Third, for individual gig workers, an increase in the quantity of labor supply will lead to a decrease in quality. Thus, gig economy platforms may not want to let individual gig workers increase their labor supply too much. Instead, gig economy platforms should balance

the amount of labor supply from different gig workers and encourage high-quality workers to provide more labor supply on the platform. They can design the payment scheme to induce more high-quality labor supply.

Last but not least, from a policy perspective, our findings imply that gig economy serves as a buffer during major social disruptions (like the pandemic), but the shock brought by the pandemic does not change people's preference in (gig economy) jobs and working styles in the long run.

7.2. Limitations and Future Research Directions

There are several limitations of our work, and some of them may point toward fruitful research directions.

First, we want to discuss the causality of our results. Our study is based on observational data, so we can only try our best to tease out alternative explanations and get causal estimates as much as we can instead of directly claiming them to be causal as based on experimental data. The before and after analyses we conduct can only estimate the change in gig workers' labor supply after the outbreak of the pandemic (compared with before the pandemic). It is not a causal approach in the sense that the change may be caused by the COVID-19 pandemic or other shocks happening in the same time period or can be caused by seasonality. The year to year DiD analysis can exclude the seasonality effects, but it cannot exclude other shocks that might happen after the outbreak of the pandemic. If we want to claim the estimated effects to be causal, we need to make an additional assumption that there are no other major shocks irrelevant to the COVID-19 pandemic happening during this period (although this assumption is largely valid based on our understanding of the context and conversations with managers at the company). When disentangling the impact of several possible driving forces, we make use of the fact that different states have different levels of pandemic severity over time and also, have different magnitudes of increase in the unemployment rate and different starting and ending dates of the selected NPIs. These variations across states and over time allow us to causally identify how much these factors can explain gig workers' change in labor supply. For future research, although we think it is infeasible to use experimental methods to get a clean causal estimate of the total impact of the pandemic on labor supply, it might be feasible to use experimental methods to provide better causal estimates of the impact of certain factors (like working at home policies) on gig workers' labor supply but likely in a more limited setting.

The relationship between gig workers' earning potential on the platform and their change in labor supply is, strictly speaking, not a causal estimate of the impact of gig workers' earning potential on their change in labor supply because gig workers' earning potential is related

to their quality, and the quality of gig workers also affects their change in labor supply. This issue cannot be fully fixed by including worker fixed effects in the regression. The relationship between quality and quantity of labor supply captured in Table A13 in the online appendix is not strictly causal either because the quantity of labor supply is time varying, and thus, worker fixed effects cannot capture the time-varying shocks that may affect both the quality and quantity. Even though these results are not strictly causal, they can still provide valuable insights; they can help gig economy platforms in understanding gig workers' behavior and designing a better incentive scheme for gig workers to provide labor supply with higher quality and stability.

Second, our findings are based on the shocks brought by the COVID-19 pandemic. As we decompose the impacts of the possible driving forces, we can possibly predict the change in gig workers' labor supply in other major social disruptions, although such generalizations should be cautious. It will be interesting to conduct a metastudy to compare our results with other studies on similar natural shocks.

Lastly, our data are from an online gig economy platform where the work is done purely online. Thus, the implications we get should be more applicable to those gig economy platforms where gig workers perform the work online, like Upwork, Catalant, Guru.com, etc. Generalizations to the gig economy platforms where the work requires physical interactions (like Uber, DoorDash, etc.) should be cautious. In particular, the NPIs that affect the labor supply on these platforms can be different from the NPIs that affect the platform in our study. Hence, it will be interesting for future researchers to understand how the labor supply to these platforms or traditional firms changes over time and compare that with our results.

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Endnotes

- ¹ See <https://www.pewresearch.org/social-trends/2020/12/09/how-the-coronavirus-outbreak-has-and-hasnt-changed-the-way-americans-work/>.
- ² See <https://www.who.int/emergencies/diseases/novel-coronavirus-2019>.
- ³ See <https://trumpwhitehouse.archives.gov/presidential-actions/proclamation-declaring-national-emergency-concerning-novel-coronavirus-disease-covid-19-outbreak/>.
- ⁴ See <https://www.crawco.com/blog/gig-economy-soars-during-covid-19-pandemic>.

- ⁵ See <https://www.forbes.com/sites/rebeccahenderson/2020/12/10/how-covid-19-has-transformed-the-gig-economy/?sh=3b480dcf6c99>.
- ⁶ See <https://www.pewresearch.org/social-trends/2020/12/09/how-the-coronavirus-outbreak-has-and-hasnt-changed-the-way-americans-work/>.
- ⁷ See <https://www.forbes.com/sites/rebeccahenderson/2020/12/10/how-covid-19-has-transformed-the-gig-economy/?sh=3b480dcf6c99>.
- ⁸ See the U.S. Bureau of Labor Statistics local area unemployment statistics for 2020 at <https://www.bls.gov/lau/#data>.
- ⁹ See the *New York Times* (<https://www.nytimes.com/interactive/2021/us/covid-cases.html>).
- ¹⁰ A worker can open any half-hour slot in the 24 hours a day, but in reality, more than 99% of the open slots are between 8 a.m. and 10 p.m. Beijing time.
- ¹¹ There can be overlapped workers in the 2019 and 2020 samples.
- ¹² See <https://www.openicpsr.org/openicpsr/project/119446>.
- ¹³ Notice that the dummy "Month 3" equals one if the date is in March and after March 13, the date of declaration of national emergency.
- ¹⁴ See <https://www.pewresearch.org/social-trends/2020/12/09/how-the-coronavirus-outbreak-has-and-hasnt-changed-the-way-americans-work/>.
- ¹⁵ See <https://www.bls.gov/opub/mlr/2021/article/unemployment-rises-in-2020-as-the-country-battles-the-covid-19-pandemic.htm>.
- ¹⁶ See <https://sgp.fas.org/crs/misc/R46554.pdf>.
- ¹⁷ See <https://www.forbes.com/sites/rebeccahenderson/2020/12/10/how-covid-19-has-transformed-the-gig-economy/?sh=3b480dcf6c99>.
- ¹⁸ Although we collected six variables to measure the severity of the COVID-19 pandemic, to keep the graph lean, we only plot the arguably most representative two (New Cases and New Deaths) in the graph, both at the national level. We will use all six measures at the state level in the regression analysis (Tables 2–4).
- ¹⁹ This is based on the assumption that besides influencing the labor supply directly, the other two factors are the only channels through which the focal mediator can influence the labor supply. We believe that this is a reasonable assumption because we do not find other major channels through which each of the three factors can influence the labor supply. If there exists such a channel, then the actual direct effect should be smaller than the direct effect estimated here.
- ²⁰ To find the best combination of the three factors that can explain most of the increase in labor supply, the most rigorous way is to test and compare all the eight possible combinations ($2 \times 2 \times 2 = 8$), which is almost impossible. Thus, we use a heuristic way here; we use either all current values or all previous values for the three factors and compare the mediation effect for each of the three factors to determine whether to use the current or previous value for this factor.
- ²¹ In Table A11 in the online appendix, we show that the selected NPIs can explain 89.2% and 91.4% of the increase in unemployment in stage 1 and stage 2, respectively.
- ²² The hourly wage that a worker earns on the platform is around \$20. For data confidentiality, we cannot reveal the actual distribution of the hourly wage, and workers' hourly wages are normalized to the standard normal distribution.
- ²³ We also show this pattern graphically in Figure A3(a) in the online appendix.
- ²⁴ More accurately speaking, by "earning potential" we mean gig workers' perceived earning potential on the platform, and we assume a gig worker's perceived earning potential on the platform is fixed within our observation period.
- ²⁵ Figure A3(b) in the online appendix shows the same pattern graphically.
- ²⁶ A worker may also fail to fulfill a booked slot because of the student's reasons. For example, the student may cancel the class, not

show up for the class, or encounter a technical issue. These situations do not belong to a worker problem.

²⁷ Notice that workers with low booking rates may not be of lower quality. They could be relatively new workers with few class records so that students are less likely to choose them.

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